



Supported Cards and Codecs

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This page shows which codecs / cards are supported by the Chameleon hardware and software.

Emulation

Card	used codec	codec support (hardware)	codec support (software)	application layer support	notes
any non 13.56 MHz tag		NO	NO	NO	
Mifare Ultralight	ISO 14443 A 106 kbit/s	yes	yes	yes	
Mifare Ultralight EV1	ISO 14443 A 106 kbit/s	yes	yes	yes	
Mifare Classic 1K/4K 4B/7B	ISO 14443 A 106 kbit/s	yes	yes	yes	
Mifare DESFire	ISO 14443 A with higher data rates	low bit rate yes, possibly also higher bit rates	only low bit rate	yes	

Card	used codec	codec support (hardware)	codec support (software)	application layer support	notes
Mifare DESFire EV1	ISO 14443 A with higher data rates	low bit rate yes, possibly also higher bit rates	only low bit rate	yes	backward compatible
Mifare DESFire EV2	ISO 14443 A with higher data rates	low bit rate yes, possibly also higher bit rates	only low bit rate	yes	backward compatible
Mifare PLUS	ISO 14443 A with higher data rates	low bit rate yes, possibly also higher bit rates	only low bit rate	no	
any NTAG	ISO 14443 A 106 kbit/s	yes	yes	no	
LEGIC prime	LEGIC prime / ISO 14443 A / ISO 15693	probably / yes / yes	no / yes / work in progress	no	
HID iCLASS	125 kHz / ISO 15693 / ISO 14443 B	no / yes / yes	no / yes / no	no	
ePass	ISO 14443 A / B	yes (possibly also higher data rates) / yes	only low bit rate / no	no	
any ISO 15693	ISO 15693	yes	yes	yes	

Sniffing

codec	codec support (hardware)	codec support (software)	application layer support	notes
any non 13.56 MHz codec	NO	NO	NO	
ISO 14443 A 106 kbit/s	PCD->PICC direction yes, possibly other direction also	currently only PCD->PICC direction	yes	
ISO 14443 A higher datarates	possibly	no	basically the same like for lower data rates	
ISO 15693	yes	VCD->VICC yes, VICC- >VCD only single subcarrier	not yet	configurable threshold in work

Reader

Card	used codec	codec support (hardware)	codec support (software)	application layer support	notes
any non 13.56 MHz tag		NO	NO	NO	
Mifare Ultralight	ISO 14443 A 106 kbit/s	yes	yes	yes	command dump_mfu
Mifare Classic 1K/4K 4B/7B	ISO 14443 A 106 kbit/s	yes	yes	no	no command for reading a MF classic card implemented yet,

Card	used codec	codec support (hardware)	codec support (software)	application layer support	notes
					cryptographic functions already implemented
Mifare DESFire	ISO 14443 A with higher data rates	low bit rate yes, possibly also higher bit rates	only low bit rate	no	no command for reading implemented yet, cryptographic functions work on progress

Notes for MIFARE DESFire emulation

There is some limited DESFire support. Please see the DESFire-specific readme [here](#). As of 1st of May 2023, the following has been tested:

- UID emulation and change
- Authentication using all available ciphers
- App creation, key creation/change when the app uses AES
- AES encrypted file operations (max file len 32 bytes)
- CMAC calculation and IV updates for most operations (AES only!)

If some functionality is not listed above, it does not mean that it will not work. It just wasn't tested. We focused on AES support first. Other algorithms may be available later. Most of the DESFire code was submitted in a PR (see the Readme). Any EV2/EV3 specific functionality is NOT available.

Experimental support for Gallagher

There's some support for pentesting Gallagher systems that use AES-encrypted cards (and AES only!). Please see the Gallagher-specific Readme [here](#). This is available as of the 1st of May 2023 and may be (re)moved at any time as space on the device is in short supply.

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Notes for MIFARE DESFire emulation

Experimental support for Gallagher

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